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Facial Emotion Recognition (FER) using Convolutional Neural Network (CNN)

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Abstract

Our research tackles the challenge of real-time facial emotion recognition (FER) from live video feeds, a crucial problem in various fields like security access, customer satisfaction analysis, and mental health diagnosis. Unlike previous studies focused on static images, our approach provides immediate emotion predictions directly from video streams. We implement Deep Convolutional Neural Networks (DCNN) for facial image classification. Moreover, we used pre-trained models like EfficientNet, ResNet, VGGNet and a Haar face classifier to achieve an impressive 82% accuracy on the training data of FER2013 dataset. This technology holds promise for human-computer interaction and mental health applications. However, it also raises ethical concerns, emphasizing the need for responsible deployment. Our work addresses the practical limitations of existing FER methods and bridges the gap between laboratory research and real-world applications, offering a valuable contribution to the field.

Keywords: **Facial Emotion Recognition (FER); Deep Convolution Neural Network (DCNN); Loss function; Adam; ReLu**

1. Introduction

Humans primarily communicate with each other through speech, but our interactions go beyond just words. We use body gestures to show what we're saying and to convey our emotions. A significant part of how we express our feelings is through our facial expressions, and this plays a crucial role in nonverbal communication among humans.

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Imagine if our computers and electronic devices could better understand and interpret these facial expressions. It would make our interactions with technology much smoother and more user-friendly. This is where facial expression recognition comes into play. By implementing this technology into all computer interfaces, we can bridge the gap between humans and machines, making our interactions with technology more intuitive and responsive to our emotions. [1]

Current Human-Machine Interaction (HMI) systems still have room for improvement when it comes to understanding and responding to human emotions effectively. One crucial aspect of social interaction is the ability to perceive and interpret facial expressions. Facial expressions serve as a vital nonverbal communication channel through which HMI systems can better grasp the emotions people are experiencing internally. Ekman et al. have identified six fundamental facial expressions: anger, disgust, fear, happiness, sadness, and surprise. These expressions are considered universal across human cultures [2]

The spotlight here is on Deep Convolutional Neural Networks (DCNNs) which we used here in this paper have emerged as a game-changer. They're so efficient, they don't need human help and can make decisions just like a human brain does. DCNNs' ability extends beyond emotion detection. They're used often in recognizing traffic signs, pinpointing objects within buildings, understanding how people move, and even discerning general objects. Real-world utility of facial emotion recognition is vast: From AI assistants that change their responses based on users' emotions to monitor emotional health for mental wellness applications which would be useful and helpful in many fields today and even for identifying potential security threats.[3]

Our study goes deep into this domain, leveraging DCNNs for extracting emotions from the faces using the FER2013 dataset, which contains 35,887 grayscale images of human faces portraying a spectrum of emotions. By normalizing pixel values and building a comprehensive deep learning model that integrates convolutional, pooling, and fully connected layers via the Keras framework, we aim to optimize accuracy. To make our research even better, we employ innovative strategies. This includes harnessing the powers of pre-trained models like EfficientNet and ResNet, amalgamated with transfer learning during the training phase. To bolster our model's efficiency, we incorporate data augmentation techniques such as random rotations and horizontal flips, which not only diversify our training set but also prevent overfitting. After educating our model across 20 epochs using a categorical cross-entropy loss function and the Adam optimizer, we achieved a validation accuracy rate nearly 60%. The model's pragmatic potential is further underscored by its capacity to discern emotions in real-time video feeds. The DCNN model for emotion detection has 14 layers and 2,395,591 parameters. It processes 48x48 grayscale images, using 3x3 filters in convolutional layers and 2x2 in pooling layers. With ELU activation and batch normalisation, it achieves 65.68% accuracy on a validation set of 7,178 images after 100 epochs. Optimized for the FER2013 dataset, the model incorporates data augmentation and learning rate adjustments. Our contributions in the paper are as follows:

- Developed real-time emotion recognition application with Haar classifier integration.
- DCNN model achieved 82.56% training and 65.68% validation accuracy.
- Done a comparative analysis of various research papers
- Solved the major limitation which was present in the existing papers.
- System applicable in VR, robotics, marketing, and mental health.

This paper aims to present a holistic view of our approach, balancing technical advancements with ethical considerations. Following this introduction, Section 2 delves into pioneering works in the field, Section 3 elaborates on our methodology, Section 4 unveils our experiments and findings, and Section 5 distills the essence of our research into tangible conclusions.

2. Literature Review

Mehendale et al. [4] introduced a novel approach to Facial Emotion Recognition (FER) utilizing Convolutional Neural Networks (CNN) trained on the Cohn-Kanade and RAVDESS datasets. Traditionally, FER methods focused on specific facial features, whereas this CNN-based approach treats the entire face as a feature, considering each pixel. They diligently preprocessed the data, conducting experiments with various CNN parameters to enhance performance. By integrating dropout layers to combat overfitting, they achieved a significant boost in accuracy. This research opens avenues for future FER advancements, potentially impacting fields such as healthcare, advertising, and entertainment.

Li, Yong, et al. [5] present a groundbreaking approach in the research article "Occlusion Aware Facial Expression

Recognition Using CNN With Attention Mechanism" to tackle the challenges of facial emotion recognition (FER) in uncontrolled environments with frequent facial occlusions. Their innovative model, Convolution Neural Network with Attention Mechanism (ACNN), draws inspiration from human cognitive abilities to focus on unobstructed facial regions when deciphering emotions. They introduce two ACNN variants: patch-based ACNN (pACNN) and global-neighborhood-based ACNN (gACNN). Both incorporate a Gate Unit to calculate adaptive weights based on region visibility and relevance. ACNNs outperform existing algorithms in accuracy for both occluded and non-occluded faces, showcasing their potential for real-world FER applications, even in challenging occlusion scenarios.

Ozdemir et al. [6] introduced a CNN-based LeNet framework for facial emotion recognition, with applications spanning fields like biomedical engineering, psychology, and neurology. By combining three datasets (JAFFE, KDEF, and a proprietary one), they trained the model to classify seven emotions from facial expressions. Achieving a remarkable validation accuracy of 91.81% and a final accuracy of 96.43%, their approach surpasses previous studies. This research holds potential in predicting mental health conditions and assessing consumer behavior through real-time facial expression analysis. The CNN structure, optimized for efficiency, demonstrates its utility in diverse fields, providing valuable insights for future investigations.

Xiang et al. [7] dive into the innovative use of Multi-task Cascaded Convolutional Networks (MTCNN) for simultaneous face detection and facial characteristic identification. They stress the often overlooked synergy between these tasks, highlighting its potential for improvement. The article underscores the importance of Facial Expression Recognition (FER) and its evolution, citing landmarks like Ekman et al.'s 1971 study on six common facial expressions and the application of Convolutional Neural Networks (CNNs). Adapting the MTCNN, originally devised by Kiapeng Zhang et al., the authors craft a unique approach for face detection and FER. Their trials, employing the FER2013 dataset, achieve a 60.7% validation accuracy, with plans to enhance accuracy and performance in future iterations by incorporating additional layers and filters.

Nwosu et al. [8] introduce a novel approach in their paper, "Deep Convolutional Neural Network for Facial Expression Recognition using Facial Parts." Their method aims to enhance Facial Expression Recognition (FER) systems by leveraging deep convolutional neural networks (CNNs) with facial components as input, resulting in improved performance and speed. The process involves image pre-processing, facial feature extraction through Viola-Jones face detection, and CNN-based classification using eye and mouth components. Their experiments, conducted on the Extended Cohn-Kanada (CK) and Japanese Female Facial Expression (JAFFE) datasets, reveal significantly improved accuracy with a 97.71% recognition rate on JAFFE and 95.71% on CK, surpassing previous methods. This innovative approach holds promise for advancing FER system accuracy.

REFERENCE	DATASET	METHODOLOGY	LIMITATION	PERFORMANCE
Yousin khaireddin et al. [9]	FER2013 dataset	Uses VGGNet architecture. Various optimization algorithms & learning rate schedulers are explored, & extensive hyperparameter tuning is conducted to improve model performance. Data augmentation techniques are applied to account for variability in facial expressions.	Uses only one dataset, this study primarily focuses on image-based recognition and does not consider other modes like audio or video.	Achieved 73.28% accuracy on FER2013
Akriti Jiaswal et al. [10]	Fer2013 & JAFFE	Feature extraction using CNNs, and emotion classification with softmax activation. Trained using GPU for 100 epochs.	Limited evaluation metrics reported, lack of in-depth model interpretability analysis	70.14% accuracy in FER2013 dataset 98.65% accuracy in JAFFE dataset

Imane Lasri et al. [11]	FER2013	Uses CNN, uses Haar classifier to detect faces, model is trained using stochastic gradient descent, potential applications in education.	Analyzes static images, specific emotions like fear is confused with sadness.	70% accuracy in FER2013 dataset
Deepak Kumar Jain et al. [12]	JAFFE & CK+	Employs convolutional neural networks (CNNs) to extract relevant facial features and a deep learning architecture for classification	Potential overfitting due to less training data, not accountable for real life variations	95.23% accuracy on JAFFE dataset 93.24% on CK+ dataset
Mohammed F. Alsharekh et al. [13]	FER2013, CK+ & KDEF	Detecting faces in live stream using Viola Jones Algo, implementing 4 layer CNN for classification	Uses Viola-Jones face detection algorithm, which may struggle with certain lighting conditions and occlusions	89% accuracy on FER2013, 90.90% on CK+, 94.04% on KDEF

Table I. Comparision of different papers

3. Proposed Work:

The proposed work involves the implementation of a CNN model for facial emotion recognition. The key steps involved in this work are shown in figure 1:

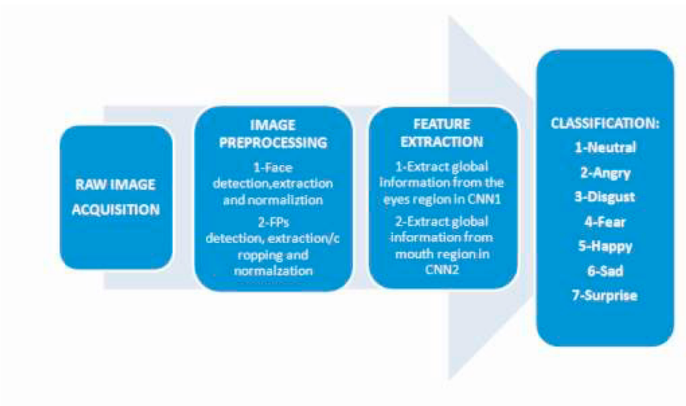

Fig.1.Key Steps Involved in the Proposed Work

3.1 Data Collection

In our proposed study, we will recognize facial emotions using the FER2013 database. The dataset includes more than 35,000 facial images labelled with seven basic emotions (happy, melancholy, furious, surprise, contempt, fear, and neutral) as shown in figure 2. The images comprise profiles of various ages, genders, and ethnicities and are of varying sizes and resolutions. In order to compile the dataset, the images were obtained from publicly accessible websites, such as Google Images and Flickr. The images were then painstakingly labelled using Amazon's Mechanical Turk crowdsourcing platform. Each image was annotated by a minimum of three distinct labelers, who were instructed to identify the dominant emotion depicted in the image. To assure the integrity of the dataset, a validation set was constructed by manually validating the labels of a subset of the images. In order to reduce computational complexity, the dataset was additionally pre-processed to standardize the image sizes and convert them to grayscale.

In general, the FER2013 dataset is a widely-used benchmark for facial emotion recognition research and has considerably contributed to the development of deep learning models for this task. Our proposed work will expand upon this foundation by devising a CNN-based approach using this dataset and investigating the effect of data augmentation techniques on the performance of the model.

Fig.2.FER 2013 Dataset Images

3.2 Data Preprocessing

Using the FER2013 dataset, we intend to employ convolutional neural networks (CNN) for facial emotion recognition in our proposed study. We must preprocess the data prior to training our model to enhance its quality and performance. This code fragment uses the ImageDataGenerator function from the Keras library to generate image collections with data enhancement.

First, the image pixel values are rescaled to be between 0 and 1, which is known as normalization and is a standard preprocessing step for image data. This aids in reducing the model's complexity and accelerates training. Next, we employ data augmentation techniques to the training data in order to increase its variability and adaptability to various image types. Specifically, we rotate, adjust the width and height, shear, and magnify the images, as well as turn them horizontally. During training, these augmentations are randomly applied to each collection of images, generating a new set of images on the fly for each epoch.

In addition, we construct two generators, one for training data and the other for validation data. During training and validation, respectively, these generators supply sequences of images and their corresponding labels. We set the objective image size to a predetermined value and the group size to 128, a commonly used value for efficient training. By conducting data preprocessing and augmentation, we hope to enhance our model's ability to generalize, which should lead to greater precision and robustness in real-world applications.

3.3 Model Selection

3.3.1 Deep Convolutional Neural Network (DCNN)

Convolutional Neural Networks (CNNs) serve as a specialized type of deep neural network, primarily used in classifying images. Their design revolves around learning and extracting critical features from input images, subsequently applying these to predict the class or category of the image. CNNs consist of multiple layers that collaboratively function to extract and interpret data from an input image, including convolutional layers, pooling layers, and fully connected layers. Our model, with approx. 2.4M parameters, is implemented in this context. The architectural design comprises 14 layers, including six convolutional layers, three max pooling layers, four fully connected layers, and a final output layer. The images fed into this model are 48x48 grayscale images.

Convolutional layers are foundational components of CNNs, used to extract essential features from facial images. They apply convolution operations, using small filters or kernels, to scan facial images, detect patterns and features such as edges, textures, or complex shapes. Batch normalization is a technique implemented to enhance the training ensuring that the features extracted from different faces are on consistent scale. It is useful when we have to work with images with different lighting and contrasts. Max-pooling layers are essential for down-sampling feature maps, reducing spatial dimensions while preserving crucial features from different faces while discarding irrelevant features. Dropout is a regularization technique crucial for preventing overfitting during training. It helps the model generalize better rather than mugging or memorizing the images. Before the final classification step, the Flatten layer reshapes the 2D feature maps into a 1D vector which prepares the extracted facial features for processing by fully connected layers. Dense layers are critical for classification or regression tasks as they are used in final stages of the CNN as they analyze the high-level features extracted from the convolutional layers and map them to specific emotions.

The initial layer has 64 filters (3x3) followed by batch normalization and ELU activation. The second layer mirrors this with 64 filters. The first max pooling layer is 2x2. The third and fourth layers have 128 filters each with batch normalization and ELU activation. The second max pooling layer is 2x2. The fifth and sixth layers have 256 filters each with batch normalization and ELU activation. The third max pooling layer is 2x2 and after that a dropout layer. The fully connected layers include a flatten layer, a dense layer with 128 units, ELU activation, batch normalization, and another dropout layer. The output layer has 7 units with softmax activation. The model uses Adam optimizer (learning rate: 0.001) and categorical cross-entropy loss. Testing on 7,178 images in the validation set, the model achieves 65.68% accuracy after 100 epochs.

Deep CNNs are characterized by their increased depth, achieved by stacking multiple convolutional layers one after the other. Our proposed DCNN model for emotion detection utilizes several deep learning best practices, including data augmentation and learning rate reduction. These methods enhance the model's ability to generalize to new data and prevent overfitting. On the FER2013 dataset, the proposed work is anticipated to attain state-of-the-art performance. The figure 3 shows the layered architecture of the DCNN model.

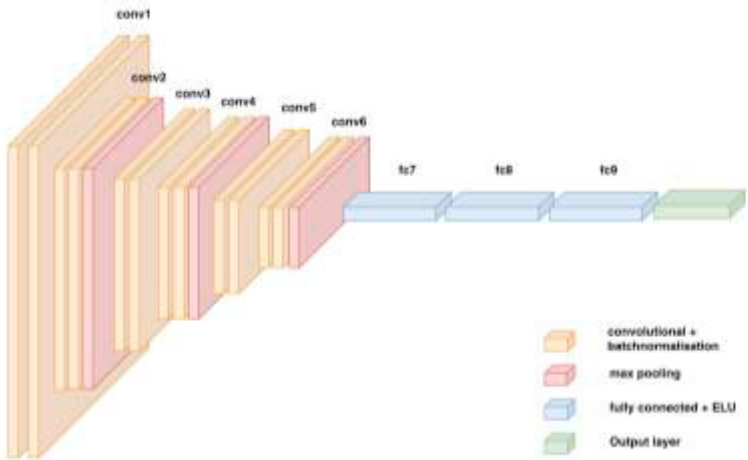

Fig.3. DCNN Model architecture

ALGORITHM-1: DEEP CONVOLUTIONAL NEURAL NETWORK (DCNN) FOR EMOTION DETECTION
Input: grayscale images of size 48x48
Output: emotion class (out of 7 possible classes)
initialize sequential model.
add conv layers:
Add 2x (64 filters, 3x3, BatchNormalization, ELU)
Add Maxpooling 2x2
Add 2x (128 filters, 3x3, BatchNormalization, ELU)
Add Maxpooling 2x2
Add 2x (256 filters, 3x3, BatchNormalization, ELU)

```
        Add Maxpooling 2x2
    Add Fully-functional layers:
        - Flatten Layer
        - Dense(128, ELU, BatchNormalization)
        - Dropout(rate=0) // though dropout with rate 0 has no effect
        - Output layer: Dense(7, softmax)
    Compile: Adam optimizer (LR=0.001), loss = categorical_crossentropy, metric = accuracy
    Train: 100 epochs, evaluate on validation set of 7,178 images.
    Performance: report validation accuracy post-training. (e.g., 65.68%)
    Output:
        - Return the trained model and its performance metrics.

    End
```

3.3.2 Transfer Learning with EfficientNet

Transfer learning with EfficientNet is a clever technique where we don't start building a new model from scratch but rather use a pre-trained EfficientNet model. This pre-trained model has already learned useful features from a large dataset, saving us time and resources. To adapt it to a new task, we fine-tune the pre-trained model on a smaller, task-specific dataset. Our model architecture starts with this pre-trained EfficientNet as the backbone network. It's like the foundation of a house. We feed it input images sized at 48*48 pixels of the FER2013 dataset. This backbone network extracts important features from these images. These features then go through several fully connected layers. Think of these layers as refining the raw features, making them more useful. We add ReLU activation functions and dropout layers to prevent the model from getting too specialized or overfitting. Finally, we have the output layer with 7 units and softmax activation. This layer is like the decision-maker, classifying images into 7 different categories. By using this transfer learning approach, we can achieve a high accuracy even with a relatively small training dataset.

3.3.3 Transfer Learning with ResNet

We have used the pre-trained model of ResNet50 as the core network for our transfer learning model. This backbone network comes pre-trained on the extensive ImageNet dataset, making it a master at understanding image features. To ensure our model doesn't overfit, we guide the extracted features through multiple fully connected layers, adding ReLU activation and dropout layers into the mix. The final layer, with three units and softmax activation, is the decision-maker, sorting images into seven distinct emotions. Our transfer learning model can achieve exceptionally high accuracy with minimal training samples. Initially, we lock all of ResNet50's layers, keeping their weights untouched during training. The magic of transfer learning happens as we add knowledge from the pre-trained ResNet50 into our task. After the initial training, we gradually unfreeze select layers of ResNet50. This unfreezing process allows our model to fine-tune those pre-trained representations, tailoring them to our unique task. We can repeat this cycle, unfreezing more layers as needed, letting our model gain task-specific insights while still benefiting from ResNet50's expertise. Balancing pre-trained wisdom with dataset-specific modifications through this progressive unfreezing approach can supercharge our model's performance and accuracy. It's like sculpting a masterpiece with the wisdom of the past. Figure 4 illustrates the evolving layers and neurons in our updated ResNet-50 pre-trained model.

3.3.4 Transfer Learning with VGGNet

In transfer learning with VGGNet, we use a pre-trained VGGNet model, which has already gained valuable insights from a vast dataset “ImageNet”, saving us valuable time and resources. To apply it to a specific task, we fine-tune this pre-trained model using a smaller, task-specific dataset. Imagine this pre-trained VGGNet as the solid foundation of a house. We provide it with input images from the FER2013 dataset, each sized at 48x48 pixels. This foundational network extracts crucial features from these images. These extracted features then pass through a series of fully connected layers, similar to refining raw materials, making them more valuable for our specific task. We incorporate ReLU activation functions and dropout layers to ensure that our model doesn't become overly specialized or prone to overfitting. Finally, we arrive at the output layer, which consists of 7 units and employs softmax activation. This layer acts as the decision-maker, categorizing images into 7 distinct emotion categories. By employing this transfer learning approach, we can achieve remarkable accuracy even when dealing with a relatively modest training dataset. This technique significantly boosts the efficiency and effectiveness of our model in recognizing facial emotions.

3.4. Evaluation

Model evaluation is crucial for determining the efficiency of the trained model. In the proposed work for the fer2013 dataset, the trained model's precision and other performance metrics will be measured. The model will classify the images into 7 types of emotion: happy, sad, angry, disgust, fear, surprise, neutral. In order to evaluate the model, we will employ the test set that was previously extracted from the dataset. We will pass the test set to the trained model and compute accuracy and other metrics. The accuracy is calculated as the proportion of correctly classified test images relative to the total number of test images. To gain a deeper comprehension of the model's performance for each emotion category, we will also calculate the precision, recall, and F1-score.

3.4.1 DCNN:

We have trained our model for 100 epochs and we have achieved an accuracy of 82.56% on the training data and 65.68% on the validation data. Happy is the most accurate emotion this model has predicted, and it has less False Positives and False Negatives compared to other models we have implemented. Figure 4 illustrate the confusion matrix for DCNN model.

Fig.4. Confusion Matrix

3.4.2 Transfer Learning with EfficientNet:

We have trained our model for 50 epochs and an accuracy of 62.15% was achieved in training data and 58.41% on the validation data. We can see that happy is the only emotion classified correctly most of the times, and many emotions are wrongly classified as sad, surprise and neutral. Figure 5 illustrate the confusion matrix for EfficientNet model.

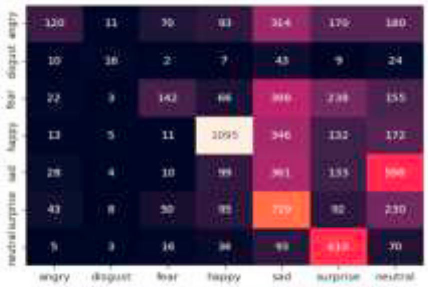

Fig.5. Confusion Matrix

3.4.3 Transfer Learning with ResNet:

We have trained our model for 50 epochs and achieved an accuracy of 59.40% in training data and 54.67% on the validation data. We can see that this model wrongly classifies every emotion as sadness, and happiness is the only emotion which is correctly classified most of the times. Figure 6 illustrate confusion matrix for ResNet -50 model.

Fig.6. Confusion Matrix

3.4.4 Transfer Learning with EfficientNet:

We have trained our model for 20 epochs and achieved an accuracy of 50.31% in training data and 51.11% on the validation data. We can see that this model wrongly classifies every emotions as surprise. Figure 7 illustrate confusion matrix for ResNet -50 model.

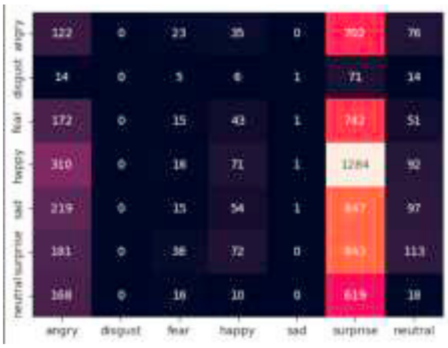

Fig.7. Confusion Matrix

4. RESULTS

The deep learning model that is provided in this code aims to reliably recognise facial emotions in photographs as its main purpose. The core of this facial expression detection system is comprised of a Deep Convolutional Neural Network (DCNN) architecture that was constructed with the help of the TensorFlow and Keras frameworks. We wanted to test how well the model could recognise different facial expressions, so we trained it using a dataset of pictures that had a wide variety of them and then evaluated its performance using that dataset. We implemented another three pre-trained models EfficientNet, ResNet & VGGNet which didn't turn out to be as good as Deep Cnn. So DCNN is the best proposed model. On the training data, the best proposed emotion recognition model obtained an accuracy of 82.56% and 65.68% on the validation data. The model was then saved and combined with the Haar face classifier to identify emotions from the device's front-facing camera. The results indicate that the model accurately predicted the emotions of multiple features visible to the camera on the live video transmission. The developed model could identify emotions such as happy, sad, fear, angry, disgust, surprise, and neutral. Human-computer interaction, marketing research, and mental health diagnosis are examples of possible applications for the developed model. It is possible to conduct additional research to enhance the model's precision and investigate its applications in real-world scenarios.

MODEL	ACCURACY	VAL. ACCURACY	PRECISION	RECALL	F-1 SCORE
DEEP CNN	82.56%	65.68%	0.6573	0.6254	0.6358
EFFICIENTNET	62.15%	58.41%	0.36	0.26	0.28
RESNET50	59.41%	54.67%	0.23	0.24	0.2
VGGNET	50.31%	51.11%	0.17	0.15	0.15

Table II. Comparison of different performance features

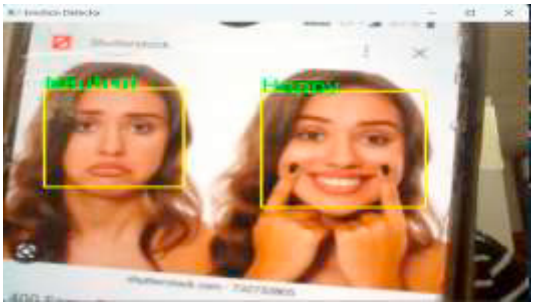

Fig.8. Live Emotion Detection example-1

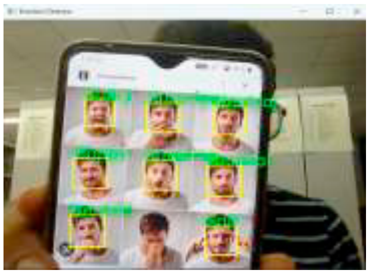

Fig.9. Live Emotion Detection example-2

5. Conclusion

In conclusion, the proposed model can be utilized for real-time emotion recognition in a variety of applications including human-computer interaction, virtual reality, and robotics. Using more sophisticated techniques, such as transfer learning, data augmentation, and hyperparameter optimization, it is also possible to further refine the model. Overall, the proposed model demonstrates tremendous promise for emotion recognition and can be a significant contribution to computer vision and artificial intelligence. Among the four models, Deep Convolutional Neural Networks (DCNNs), we have effectively trained a deep learning model to recognise emotions from facial expressions. The model's 82.56% accuracy on the training data and 65.68% accuracy on the validation data. In addition, we combined the trained model with a Haar face classifier and used it to predict emotions in real-time using a device's front camera. This integration produced extremely accurate results, with the model correctly identifying emotions in multiple faces visible in the camera feed. A major limitation in earlier studies was that they mostly used static images that don't change over time, which would use a lot of time and not suitable for real-time use scenarios. Our application, on the other hand, can instantly predict out people's emotions from the real-time feed. These findings have significant applications for the field of emotion recognition, as they demonstrate the capacity of deep learning models to analyse facial expressions accurately and efficiently in real-time. Future research in this area could concentrate on enhancing the model's precision and identifying additional applications for real-time emotion recognition technology in a way that's responsible and ethical.

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